

StatsChitran: Mathematical and Practical Documentation

Chitran Ghosal

Contents

1	Introduction	1
2	Linear modelling functions	2
2.1	The linreg function	2
2.1.1	The univariate function	3
2.1.2	The multivariate function	4
2.1.3	Matrix representation	7
2.1.4	Function Implementation	8
2.1.5	Examples	9

1 Introduction

This vignette provides mathematical and practical documentation for selected functions in the **StatsChitran** package.

The aim is to document both the usage of the functions and the mathematical reasoning behind their implementation.

2 Linear modelling functions

2.1 The linreg function

This function `StatsChitran::linreg` involves fitting a linear regression model to a multivariate case while returning a message in case of multicollinearity

The model used is a non-weighted OLS

2.1.1 The univariate function

Start with a simple fit for a $y = ax + b + \epsilon$ case where $\epsilon \sim N(0, \sigma^2)$. Obviously the model in this case would be $f(x) = ax + b$

Then let $(x_i, y_i) \mid i \in [1, n]$ be the dataset. Then the the objective cost function involving the RSS according to OLS becomes

$$\mathcal{L}(a, b) = \sum_i \{y_i - ax_i - b\}^2 \quad (1)$$

$$(2)$$

For minimizing $\mathcal{L}(a, b)$, we need

$$\nabla \mathcal{L} = \begin{pmatrix} \partial \mathcal{L} / \partial a \\ \partial \mathcal{L} / \partial b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Hence,

$$\frac{\partial \mathcal{L}}{\partial a} = -2 \sum_i \{y_i - ax_i - b\} x_i = 0 \quad (3)$$

$$\implies a \sum_i x_i^2 + b \sum_i x_i = \sum_i x_i y_i \quad (4)$$

Additionally,

$$\frac{\partial \mathcal{L}}{\partial b} = -2 \sum_i \{y_i - ax_i - b\} = 0 \quad (5)$$

$$\implies a \sum_i x_i + b \sum_i 1 = \sum_i y_i \quad (6)$$

Thus, in this case, estimating the coefficient pair a, b is a simple case for solving for the matrix

$$\begin{pmatrix} \sum_i x_i^2 & \sum_i x_i \\ \sum_i x_i & \sum_i 1 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \sum_i x_i y_i \\ \sum_i y_i \end{pmatrix}$$

Avoiding multicollinearity simply becomes a restriction of imposed non singularity on the matrix

$$M = \begin{pmatrix} \sum_i x_i^2 & \sum_i x_i \\ \sum_i x_i & \sum_i 1 \end{pmatrix}$$

In a strict theoretial sense that becomes

$$\sum_i x_i^2 \neq \frac{1}{n} \left\{ \sum_i x_i \right\}^2$$

That is only possible if all x_i values are the same, hence the dataset would be reduced by a dimension of at least one with respect to the geometry that it is trying to fit itself to, basically the definition of a singular matrix.

2.1.2 The multivariate function

Now, if the issue is extended to the multivariate case, herein one would have $y = a_1x_1 + a_2x_2 + \cdots + a_mx_m + a_{m+1} + \epsilon$ with $\epsilon \sim N(0, \sigma^2)$. Obviously the linear fit in this case is $f(x) = a_1x_1 + a_2x_2 + \cdots + a_mx_m + a_{m+1}$.

Thus the dataset in this case would be $(x_{1,i}, x_{2,i}, \cdots, x_{m,i}, y_i)$ with $i \in [1, n]$

In this case the objective would be $\mathcal{L}(a_1, a_2, \cdots, a_{m+1})$

$$\mathcal{L} = \sum_i \{a_1x_{1,i} + a_2x_{2,i} + \cdots + a_mx_{m,i} + a_{m+1} - y_i\}^2 \quad (7)$$

$$= \sum_{i=1}^n \left\{ a_{m+1} - y_i + \sum_{k=1}^m a_k x_{k,i} \right\}^2 \quad (8)$$

Thus for the gradient to be zero

$$\nabla \mathcal{L} = \begin{pmatrix} \partial \mathcal{L} / \partial a_1 \\ \partial \mathcal{L} / \partial a_2 \\ \vdots \\ \partial \mathcal{L} / \partial a_m \\ \partial \mathcal{L} / \partial a_{m+1} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix}$$

That implies that $\forall p \in [1, m]$

$$\partial \mathcal{L} / \partial a_p = 2 \sum_{i=1}^n \{a_{m+1} - y_i + \sum_{k=1}^m a_k x_{k,i}\} x_{p,i} = 0 \quad (9)$$

$$\implies a_{m+1} \sum_{i=1}^n x_{p,i} + \sum_{i=1}^n x_{p,i} \left\{ \sum_{k=1}^m a_k x_{k,i} \right\} = \sum_{i=1}^n x_{p,i} y_i \quad (10)$$

$$\implies \sum_{k=1}^m a_k \sum_{i=1}^n x_{k,i} x_{p,i} + a_{m+1} \sum_{i=1}^n x_{p,i} = \sum_{i=1}^n x_{p,i} y_i \quad (11)$$

This means that the first m equations can be written down as shown below

$$a_1 \left(\sum_{i=1}^n x_{1,i} x_{1,i} \right) + a_2 \left(\sum_{i=1}^n x_{2,i} x_{1,i} \right) + \cdots + a_m \left(\sum_{i=1}^n x_{m,i} x_{1,i} \right) + a_{m+1} \sum_{i=1}^n x_{1,i} = \sum_{i=1}^n x_{1,i} y_i \quad (12)$$

$$a_1 \left(\sum_{i=1}^n x_{1,i} x_{2,i} \right) + a_2 \left(\sum_{i=1}^n x_{2,i} x_{2,i} \right) + \cdots + a_m \left(\sum_{i=1}^n x_{m,i} x_{2,i} \right) + a_{m+1} \sum_{i=1}^n x_{2,i} = \sum_{i=1}^n x_{2,i} y_i \quad (13)$$

$$a_1 \left(\sum_{i=1}^n x_{1,i} x_{3,i} \right) + a_2 \left(\sum_{i=1}^n x_{2,i} x_{3,i} \right) + \cdots + a_m \left(\sum_{i=1}^n x_{m,i} x_{3,i} \right) + a_{m+1} \sum_{i=1}^n x_{3,i} = \sum_{i=1}^n x_{3,i} y_i \quad (14)$$

$$a_1 \left(\sum_{i=1}^n x_{1,i} x_{4,i} \right) + a_2 \left(\sum_{i=1}^n x_{2,i} x_{4,i} \right) + \cdots + a_m \left(\sum_{i=1}^n x_{m,i} x_{4,i} \right) + a_{m+1} \sum_{i=1}^n x_{4,i} = \sum_{i=1}^n x_{4,i} y_i \quad (15)$$

$$a_1 \left(\sum_{i=1}^n x_{1,i} x_{5,i} \right) + a_2 \left(\sum_{i=1}^n x_{2,i} x_{5,i} \right) + \cdots + a_m \left(\sum_{i=1}^n x_{m,i} x_{5,i} \right) + a_{m+1} \sum_{i=1}^n x_{5,i} = \sum_{i=1}^n x_{5,i} y_i \quad (16)$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad = \quad \vdots \quad (17)$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad = \quad \vdots \quad (18)$$

$$a_1 \left(\sum_{i=1}^n x_{1,i} x_{m,i} \right) + a_2 \left(\sum_{i=1}^n x_{2,i} x_{m,i} \right) + \cdots + a_m \left(\sum_{i=1}^n x_{m,i} x_{m,i} \right) + a_{m+1} \sum_{i=1}^n x_{m,i} = \sum_{i=1}^n x_{m,i} y_i \quad (19)$$

$$(20)$$

However since in parameter space there are $m+1$ parameters $a_1, a_2, a_3, \dots, a_m, a_{m+1}$ hence $m+1$ conditions are needed wherein the $(m+1)$ th condition comes from setting the last minima $\partial \mathcal{L} / \partial a_{m+1} = 0$ as shown below

$$\partial \mathcal{L} / \partial a_{m+1} = 2 \sum_{i=1}^n \left\{ a_{m+1} - y_i + \sum_{k=1}^m a_k x_{k,i} \right\} = 0 \quad (21)$$

$$\implies \sum_{k=1}^m a_k \sum_{i=1}^n x_{k,i} + a_{m+1} \sum_{i=1}^n 1 = \sum_{i=1}^n y_i \quad (22)$$

In an expanded form this expands to the form shown below

$$a_1 \sum_{i=1}^n x_{1,i} + a_2 \sum_{i=1}^n x_{2,i} + \cdots + a_m \sum_{i=1}^n x_{m,i} + a_{m+1} \sum_{i=1}^n 1 = \sum_{i=1}^n y_i$$

With $m+1$ linear conditions in the parameters from a_1 to a_{m+1} , the $(m+1) \times (m+1)$ matrix representation of $[M]$ can be shown below

$$[M] = \begin{pmatrix} \sum_{i=1}^n x_{1,i}^2 & \sum_{i=1}^n x_{2,i} x_{1,i} & \cdots & \sum_{i=1}^n x_{m,i} x_{1,i} & \sum_{i=1}^n x_{1,i} \\ \sum_{i=1}^n x_{1,i} x_{2,i} & \sum_{i=1}^n x_{2,i}^2 & \cdots & \sum_{i=1}^n x_{m,i} x_{2,i} & \sum_{i=1}^n x_{2,i} \\ \sum_{i=1}^n x_{1,i} x_{3,i} & \sum_{i=1}^n x_{2,i} x_{3,i} & \cdots & \sum_{i=1}^n x_{m,i} x_{3,i} & \sum_{i=1}^n x_{3,i} \\ \sum_{i=1}^n x_{1,i} x_{4,i} & \sum_{i=1}^n x_{2,i} x_{4,i} & \cdots & \sum_{i=1}^n x_{m,i} x_{4,i} & \sum_{i=1}^n x_{4,i} \\ \sum_{i=1}^n x_{1,i} x_{5,i} & \sum_{i=1}^n x_{2,i} x_{5,i} & \cdots & \sum_{i=1}^n x_{m,i} x_{5,i} & \sum_{i=1}^n x_{5,i} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \sum_{i=1}^n x_{1,i} x_{m,i} & \sum_{i=1}^n x_{2,i} x_{m,i} & \cdots & \sum_{i=1}^n x_{m,i}^2 & \sum_{i=1}^n x_{m,i} \\ \sum_{i=1}^n x_{1,i} & \sum_{i=1}^n x_{2,i} & \cdots & \sum_{i=1}^n x_{m,i} & \sum_{i=1}^n 1 \end{pmatrix}$$

Similarly, the $(m+1) \times 1$ coefficient matrix $[C]$ can be drawn as shown below.

$$[C] = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_m \\ a_{m+1} \end{pmatrix}$$

To be able to complete the definition, the result vector also has to be defined as a $(m + 1) \times 1$ matrix $[B]$ as shown below

$$[B] = \begin{pmatrix} \sum_{i=1}^n x_{1,i} y_i \\ \sum_{i=1}^n x_{2,i} y_i \\ \sum_{i=1}^n x_{3,i} y_i \\ \vdots \\ \sum_{i=1}^n x_{m,i} y_i \\ \sum_{i=1}^n y_i \end{pmatrix}$$

With these definitions structured, now the relationship can be established

$$[M] \cdot [C] = [B]$$

Obviously the non-singularity of $[M]$ is necessary in order not to reduce its rank else the situation of multicollinearity can arise.

2.1.3 Matrix representation

Additionally, a better representation of the matrixes $[M]$ and $[B]$ can be created. For this purpose, each datacolumn which signify the independent variables $x_1, x_2, \dots, x_m$ can be treated as vectors $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_m$. An additional vector $\mathbf{1}$ is defined in this regard.

Each of these vectors are treated as possessing a length of n since the data are n -observations long as defined previously.

Thus the vectors $\mathbf{X}_r, \mathbf{Y}, \mathbf{1}$ are defined in the manner shown below.

$$\mathbf{X}_r = \begin{pmatrix} x_{r,1} \\ x_{r,2} \\ x_{r,3} \\ \vdots \\ x_{r,n} \end{pmatrix} \quad \mathbf{Y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{pmatrix} \quad \mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}$$

As a consequence of these definitions,

$$\mathbf{X}_p \cdot \mathbf{X}_q = \sum_{i=1}^n x_{p,i} x_{q,i} \quad (23)$$

$$\mathbf{X}_p \cdot \mathbf{Y} = \sum_{i=1}^n x_{p,i} y_i \quad (24)$$

$$\mathbf{X}_p \cdot \mathbf{1} = \sum_{i=1}^n x_{p,i} \quad (25)$$

$$\mathbf{Y} \cdot \mathbf{1} = \sum_{i=1}^n y_i \quad (26)$$

$$\mathbf{1} \cdot \mathbf{1} = n \quad (27)$$

$$(28)$$

Hence, the matrix definitions can be updated as shown

$$[M] = \begin{pmatrix} \mathbf{X}_1 \cdot \mathbf{X}_1 & \mathbf{X}_1 \cdot \mathbf{X}_2 & \cdots & \mathbf{X}_1 \cdot \mathbf{X}_m & \mathbf{X}_1 \cdot \mathbf{1} \\ \mathbf{X}_2 \cdot \mathbf{X}_1 & \mathbf{X}_2 \cdot \mathbf{X}_2 & \cdots & \mathbf{X}_2 \cdot \mathbf{X}_m & \mathbf{X}_2 \cdot \mathbf{1} \\ \mathbf{X}_3 \cdot \mathbf{X}_1 & \mathbf{X}_3 \cdot \mathbf{X}_2 & \cdots & \mathbf{X}_3 \cdot \mathbf{X}_m & \mathbf{X}_3 \cdot \mathbf{1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{X}_m \cdot \mathbf{X}_1 & \mathbf{X}_m \cdot \mathbf{X}_2 & \cdots & \mathbf{X}_m \cdot \mathbf{X}_m & \mathbf{X}_m \cdot \mathbf{1} \\ \mathbf{1} \cdot \mathbf{X}_1 & \mathbf{1} \cdot \mathbf{X}_2 & \cdots & \mathbf{1} \cdot \mathbf{X}_m & \mathbf{1} \cdot \mathbf{1} \end{pmatrix}$$

$$[B] = \begin{pmatrix} \mathbf{X}_1 \cdot \mathbf{1} \\ \mathbf{X}_2 \cdot \mathbf{1} \\ \mathbf{X}_3 \cdot \mathbf{1} \\ \vdots \\ \mathbf{X}_m \cdot \mathbf{1} \\ \mathbf{Y} \cdot \mathbf{1} \end{pmatrix}$$

$$[C] = [M]^{-1}[B]$$

2.1.4 Function Implementation

For the function's implementation in **R**, the function accepts a dataframe **df** as its argument. This occurs such that each individual column of **df** is each of the vectors $\mathbf{X}_p$ along with $\mathbf{Y}$ being the last column of the dataframe.

Hence,

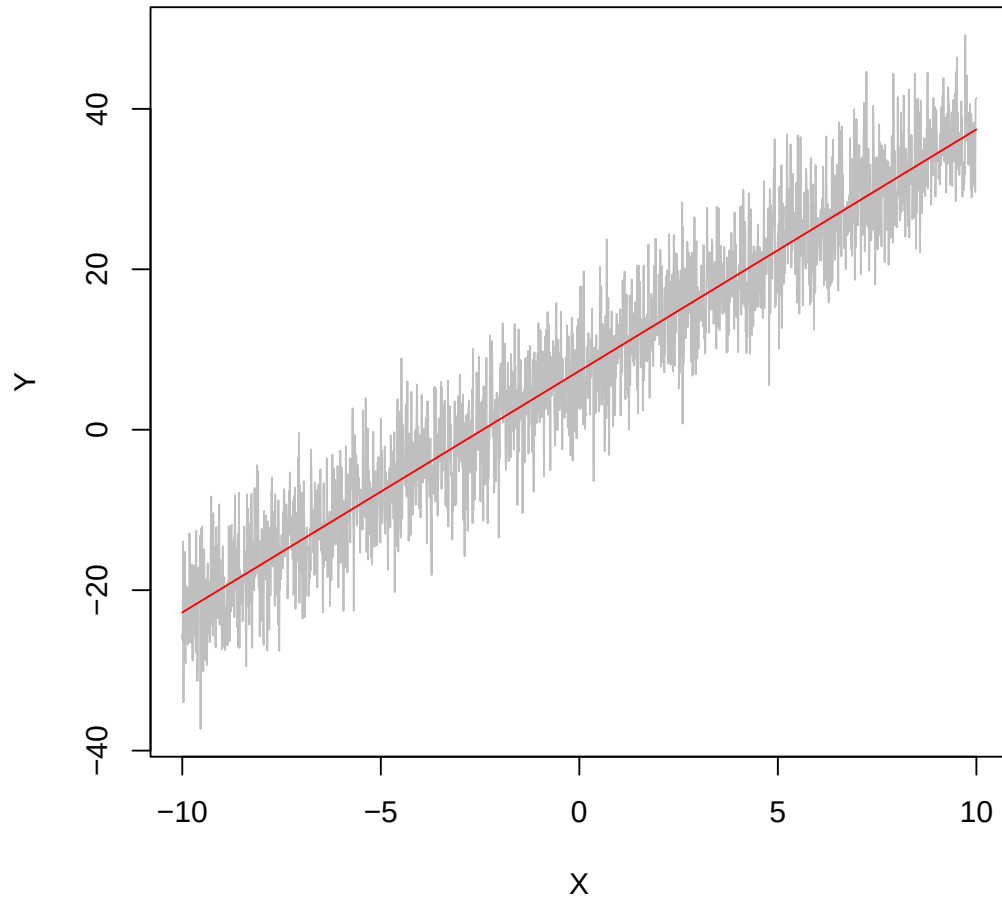
```
df[,1] =  $\mathbf{X}_1$   
df[,2] =  $\mathbf{X}_2$   
df[,3] =  $\mathbf{X}_3$   
df[,m] =  $\mathbf{X}_m$   
df[,m+1] =  $\mathbf{Y}$ 
```

Thus, the argument **df** should be a $m + 1$ column dataframe while the function returns vector $[C]^T$.

2.1.5 Examples

```
library(StatsChitran) #Call the relevant library

###Estimate and plot the univariate case
X <- seq(-10, 10, by=0.01)
Y <- 3*X + 7.47 + rnorm(length(X), sd = 5) #Add gaussian noise
df <- data.frame(X, Y) #Create the raw dataframe
coeff <- linreg(df) #Call the function
plot(X,Y, type = 'l', col=rgb(0,0,0,0.25)) #Plot the raw data first
lines(X, coeff[1]*X+coeff[2], col='red') #Plot the fit onto the data
```



```
coeff #Check the coefficients
#> [1] 3.010539 7.329396
```

```

library(StatsChitran) #Call the relevant library
###Estimate the multivariate case
##Build the dataset#####
X1 <- seq(-10, 10, by=0.01)
X2 <- X1^2
X3 <- X2^1.5
X4 <- X3^2
Y <- 7.3*X1 + 4.7*X2 + 3*X3 + 1.3*X4 + 11 + rnorm(n=length(X1), sd = 50) #Add gaussian noise
df <- data.frame(X1, X2, X3, X4, Y)
##Build the dataset#####
coeff <- linreg(df) #call the function
coeff #check the coefficients
#> [1] 7.117583 4.581439 3.014202 1.299995 11.739851

```